

Meta-Learning-Driven Joint Power Allocation and Phase-Shift Design for RIS-Assisted NOMA-ISAC Systems

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Abstract—This letter investigates a reconfigurable intelligent surface (RIS)-assisted multiple-input single-output (MISO) integrated sensing and communication (ISAC) system, where a base station (BS) simultaneously serves a near and a far user via non-orthogonal multiple access (NOMA) while illuminating a sensing target. We maximize a weighted communication-and-sensing throughput by jointly optimizing the NOMA power-allocation coefficients and the RIS phase shifts, subject to an aggregate communication QoS floor, individual near/far-user QoS floors, and a sensing QoS constraint. To tackle the resulting non-convex problem, a model-agnostic meta-learning (MAML)-based deep-learning algorithm is proposed. A lightweight network predicts the low-dimensional power split, while an inner optimizer adapts the high-dimensional RIS phase vector for each channel realization. A first-order meta-update then trains the power map using the phase-adapted channels, avoiding the unstable higher-order derivatives observed in this high-SINR setting. Simulation results show stable convergence, a strong RIS-aperture gain, and consistent improvements against OMA, no-RIS, and partially optimized baselines.

Index Terms—Reconfigurable intelligent surface (RIS), integrated sensing and communication (ISAC), non-orthogonal multiple access (NOMA), deep learning, meta-learning (MAML), MISO.

I. INTRODUCTION

INTEGRATED sensing and communication (ISAC) allows sensing and data transmission to share spectrum, antennas, and RF hardware [1]. Power-domain NOMA improves spectral efficiency by superposing users and using successive interference cancellation (SIC) [14], [12], while RISs can reconfigure blocked or weak propagation paths with programmable passive elements [13], [3].

RIS-assisted ISAC has been studied through joint active/passive beamforming [4], CRB-oriented block-coordinate optimization [5], and RIS-assisted rate-splitting [6]. For RIS-NOMA, [7] showed that learning/meta-learning can jointly adapt resource allocation and RIS phases. More generally, MAML learns an initialization that can adapt rapidly to a new task [9], [15], which is attractive here because the three power coefficients have reusable structure across channels whereas the N RIS phases are strongly channel-specific.

The contributions are threefold: 1) we formulate joint NOMA power and RIS-phase optimization with cluster, per-user, and

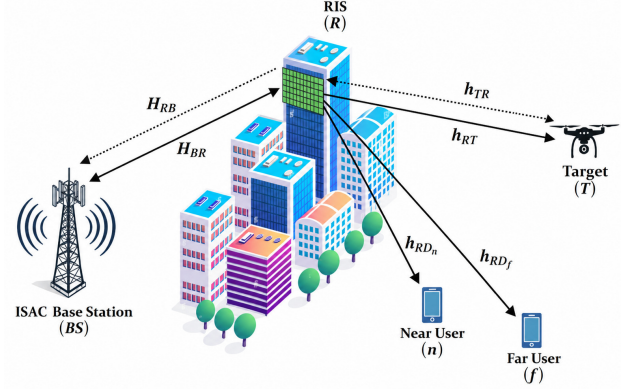


Fig. 1. RIS-assisted MISO NOMA-ISAC system and channel notation.

sensing QoS constraints; 2) we develop a first-order MAML bilevel solver in which a compact network predicts the power split and a short inner loop adapts the RIS phases; and 3) we analyze convergence limitations and complexity and compare RIS/no-RIS and NOMA/OMA designs on held-out channels.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

As shown in Fig. 1, an M -antenna BS serves near/far users and a sensing target through an N -element RIS. Element i is modeled as

$$v_i = \rho_i e^{j\phi_i}, \quad 0 \leq \rho_i \leq 1, \quad \phi_i \in [0, 2\pi), \quad (1)$$

with $\Phi = \text{diag}(v_1, \dots, v_N)$. The ideal passive-RIS setting used in simulation is given in Table I.

1) *Channel Model*: Let $\mathbf{H}_{BR} \in \mathbb{C}^{N \times M}$ denote the BS–RIS channel and let $\mathbf{H}_{RB}$ denote the reciprocal RIS–BS channel. For user $k \in \{n, f\}$, $\mathbf{h}_{RD_k} \in \mathbb{C}^N$ denotes the RIS–user channel. The effective user channel is

$$\mathbf{h}_k = \mathbf{H}_{RB} \Phi \mathbf{h}_{RD_k} \in \mathbb{C}^M. \quad (2)$$

Direct BS–user paths are blocked in the RIS-assisted data set. With $\mathbf{h}_{RT}$ and $\mathbf{h}_{TR}$ denoting the RIS–target and target–RIS links, the rank-one round-trip response is

$$\mathbf{G}_T = \sqrt{\beta_T} (\mathbf{H}_{RB} \Phi \mathbf{h}_{TR}) (\mathbf{h}_{RT}^H \Phi \mathbf{H}_{BR}), \quad (3)$$

where β_T is the target/path-gain coefficient; all numerical channel settings are listed in Section IV.

2) *NOMA Communication and Sensing Model*: The transmitted signal is

$$\mathbf{x} = \left(\sqrt{a_n P} s_n + \sqrt{a_f P} s_f \right) \mathbf{w}_c + \sqrt{a_T P} \mathbf{w}_T s_T, \quad (4)$$

where $\mathbf{a} = [a_n, a_f, a_T]$ is a nonnegative power split with $a_n + a_f + a_T = 1$. The unit-norm communication beam is

$$\mathbf{w}_c = \frac{\mu_f \mathbf{h}_f + \mu_n \mathbf{h}_n}{\|\mu_f \mathbf{h}_f + \mu_n \mathbf{h}_n\|}, \quad \mu_f, \mu_n \geq 0, \quad \mu_f + \mu_n = 1, \quad (5)$$

where μ_f and μ_n control the far/near steering emphasis; both users share $\mathbf{w}_c$ and are separated by SIC [14], [7].

For sensing, define $\mathbf{H}_u = [\mathbf{h}_n, \mathbf{h}_f]$ and the leakage-aware matrix

$$\mathbf{A}_T = \mathbf{G}_T^H \mathbf{G}_T - \lambda_L \mathbf{H}_u \mathbf{H}_u^H. \quad (6)$$

The sensing beam is the normalized dominant eigenvector of $\mathbf{A}_T$; hence λ_L controls user leakage [4], [5].

With $g_{kj} = |\mathbf{h}_k^H \mathbf{w}_j|^2$, the SINRs used by the implementation are

$$\gamma_f = \frac{a_f g_{ff} P}{a_n g_{ff} P + a_T g_{fT} P + \sigma^2}, \quad (7)$$

$$\gamma_n = \frac{a_n g_{nn} P}{a_T g_{nT} P + \sigma^2}, \quad (8)$$

$$\gamma_s = \frac{a_T |\mathbf{u}_T^H \mathbf{G}_T \mathbf{w}_T|^2 P}{\sigma^2}, \quad (9)$$

where σ^2 is noise power, $\mathbf{u}_T$ is the sensing combiner, and $R_k = \log_2(1 + \gamma_k)$. The weighted objective is

$$R_{DL} = \omega_c (R_n + R_f) + \omega_s R_s, \quad \omega_c + \omega_s = 1. \quad (10)$$

Here $R_n + R_f$ is the NOMA-cluster spectral efficiency and ω_c, ω_s set the communication/sensing priority.

B. Problem Formulation

We jointly optimize the power split $\mathbf{a}$ and RIS phases $\boldsymbol{\phi} = [\phi_1, \dots, \phi_N]$. The communication sum rate is retained as a cluster-level QoS measure, while individual user constraints are added in the same spirit as the per-device QoS constraint in (9b) of [7]. The resulting problem is

$$(P1) \quad \max_{\mathbf{a}, \boldsymbol{\phi}} \quad R_{DL} \quad (11a)$$

$$\text{s.t.} \quad R_n + R_f \geq R_{th,c}, \quad (11b)$$

$$R_k \geq R_{th,k}, \quad k \in \{n, f\}, \quad (11c)$$

$$R_s \geq R_{th,s}, \quad (11d)$$

$$a_n + a_f + a_T = 1, \quad a_n, a_f, a_T \geq 0, \quad (11e)$$

$$|e^{j\phi_i}| = 1, \quad i = 1, \dots, N. \quad (11f)$$

Constraint (11b) preserves cluster throughput, while (11c) prevents a high-rate user from masking starvation of the other, consistent with per-device QoS modeling in [7]. The sensing floor is imposed separately by (11d); the phase parameterization enforces unit modulus automatically.

Problem (P1) is non-convex because power and RIS phases are coupled. We enforce QoS through penalties, while the softmax output satisfies the power simplex.

III. PROPOSED MAML-BASED SOLUTION

A. Motivation for Meta-Learning

The three power coefficients are low dimensional and reusable across related channels, whereas the N RIS phases require channel-specific adaptation. Rather than solving the full non-convex problem from scratch or learning a one-shot high-dimensional phase map, MAML learns a reusable outer model and adapts the RIS with a fixed number of inner steps [9], [15]. This follows RIS-NOMA meta-learning [7]; the implemented first-order variant avoids the full higher-order graph [10], [11].

B. Feature Map and Power-Allocation Network

The nine-dimensional feature $\mathbf{z}$ contains six effective gains (in dB), communication/sensing QoS levels, and the SNR budget. A 9–256–256–256–3 LayerNorm/LeakyReLU network produces

$$\mathbf{a} = \text{softmax}(f_\theta(\mathbf{z})), \quad (12)$$

which enforces $a_n + a_f + a_T = 1$ and $a_k \geq 0$ by construction.

C. Joint Power-Allocation and Phase Adaptation

For a channel sample and the current phase vector $\boldsymbol{\phi}^{(j)}$, the network first predicts

$$\mathbf{a}^{(j)} = \text{softmax}\left(f_\theta(\mathbf{z}(\boldsymbol{\phi}^{(j)}))\right). \quad (13)$$

The implemented constrained loss is

$$\begin{aligned} \ell^{(j)} = & -R_{DL} + \lambda_c (R_{th,c} - R_n - R_f)_+ + \lambda_s (R_{th,s} - R_s)_+ \\ & + \lambda_n (R_{th,n} - R_n)_+ + \lambda_f (R_{th,f} - R_f)_+, \end{aligned} \quad (14)$$

where $(x)_+ = \max(x, 0)$. Here $\lambda_c, \lambda_s, \lambda_n, \lambda_f$ weight the cluster, sensing, near-user, and far-user QoS penalties. With inner index j , K phase steps, inner rate η_ϕ , outer rate α , batch size B , and network parameters $\boldsymbol{\theta}$, the updates are as follows.

Using $\theta_i = e^{j\phi_i}$ enforces unit modulus. Starting from a random phase vector, Adam performs K inner updates on (14),

$$\boldsymbol{\phi}^{(j+1)} = \text{Adam}_\phi\left(\boldsymbol{\phi}^{(j)}, \nabla_{\boldsymbol{\phi}^{(j)}} \ell^{(j)}; \eta_\phi\right), \quad (15)$$

and the features/power split are recomputed after each phase update.

D. First-Order Meta-Update

After K inner steps, the first-order MAML update treats $\boldsymbol{\phi}^{(K)}$ as fixed and updates only the power network:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha \nabla_\theta \frac{1}{B} \sum_{b=1}^B \ell_b(\boldsymbol{\theta}, \boldsymbol{\phi}_b^{(K)}), \quad (16)$$

The gradient is evaluated on phase-adapted channels; omitting derivatives through the inner optimizer reduces memory and avoids the non-finite higher-order meta-gradients observed in the implemented high-SINR setting.

E. Training and Prediction Procedure

Algorithm 1 summarizes training. At inference, the same K phase steps are applied to an unseen channel before the final softmax power split.

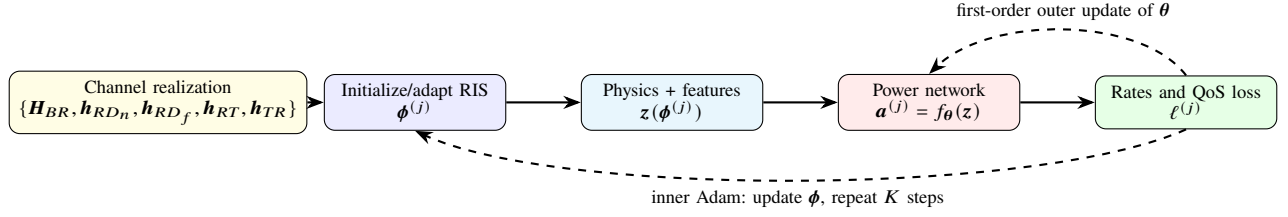


Fig. 2. First-order MAML workflow: channel-specific RIS adaptation in the inner loop and power-network training in the outer loop.

Algorithm 1: First-order MAML-based power and RIS-phase optimization

Input: channels, K , η_ϕ , α , QoS penalties.
Output: θ^* .
Initialize θ
while training **do**
 Sample B channels
 foreach channel b **do**
 Initialize $\phi_b^{(0)}$
 for $j = 0$ **to** $K - 1$ **do**
 Compute gains/features, predict $\mathbf{a}_b^{(j)}$, evaluate $\ell_b^{(j)}$
 Update $\phi_b^{(j+1)}$ by inner Adam
 end
 end
 Update θ using the first-order gradient in (16)
end
Select the best validation checkpoint

TABLE I
SYSTEM AND CHANNEL PARAMETERS.

Parameter	Value
BS antennas M	4
RIS elements N	8; sweep {8, 16, 32, 64}
BS location	[0, 0, 0] m
RIS location	[8, 0, 0] m
Near-user location	[9.69, 2.48, 0] m
Far-user location	[4.00, 24.68, 0] m
Target location	[8.00, 6.00, 0] m
Reference gain PL_0	10^{-3} (−30 dB)
Used path-loss exponents	$\alpha_{BR} = 2.2$, $\alpha_{RU} = 2.8$, $\alpha_{RT} = 2.3$
Nakagami fading classes	user-facing: $m = 1$; infrastructure/sensing: $m = 2-3$
Target coefficient β_T	40 dB
Ideal RIS amplitude ρ_i	1
Bandwidth / T / NF	1 MHz / 290 K / 6 dB
Nominal generator power	40 dBm
Power sweep	5, 10, 15, 20, 25 dBm

F. Convergence Analysis

A global guarantee is not claimed because (P1) is non-convex, the hinge QoS penalties are non-smooth at their boundaries, and the inner/outer variables differ; standard smooth MAML results therefore do not directly apply [7], [10], [11]. Convergence is assessed empirically from the fixed-step inner loop and the outer loss/rate plateau.

G. Computational Complexity

Let P_θ be the network parameter count ($\approx 1.36 \times 10^5$). One inner step requires the network forward/backward cost plus cascaded-channel and sensing-beam operations:

$$O(P_\theta + MN + M^3), \quad (17)$$

and online prediction for one channel realization costs

$$O(K(P_\theta + MN + M^3)), \quad (18)$$

and one training iteration scales as $O(BK(P_\theta + MN + M^3))$. First-order MAML avoids storing the full second-order graph through the K inner steps.

IV. NUMERICAL RESULTS AND ANALYSIS

The simulator specifies link lengths, not unique absolute coordinates. Table I therefore reports one equivalent $[x, y, z]$ embedding that reproduces those lengths; only link distances enter the path-loss model. Table II lists the learning settings.

The NOMA thresholds in Table II are those used by the fair training scripts; the OMA data use $R_{th,c} = 12$ and $R_{th,s} = 2$ bit/s/Hz. Model selection uses held-out validation performance.

TABLE II
LEARNING AND OPTIMIZATION PARAMETERS.

Parameter	Value
Data split	90% train / 10% val.; independent test
Power network	9–256–256–256–3 + softmax
Outer optimizer	Adam
NOMA-RIS training	batch 64, lr 10^{-4} , 40 epochs
Inner phase loop	$K = 10$, Adam, $\eta_\phi = 0.1$
Beam weights (μ_f, μ_n)	(0.7, 0.3) in simulations
Leakage parameter λ_L	0.5 in simulations
Objective weights (ω_c, ω_s)	(0.7, 0.3)
Cluster/sensing QoS	$R_{th,c} = 21.7$, $R_{th,s} = 2.9$ bit/s/Hz
Individual user floors	$R_{th,n} = R_{th,f} = 1$ bit/s/Hz
Penalty weights $(\lambda_c, \lambda_s, \lambda_n, \lambda_f)$	(2, 50, 15, 20)

Fig. 3 shows stable first-order training: the loss decreases and the weighted rate reaches a steady plateau across the tested settings.

Figs. 4–5 show the ablation behavior. Joint phase/power optimization raises the median weighted rate from about 5.0 to 6.7 bps/Hz and reduces joint QoS violations from 29.4% to 0.1%.

Fig. 6 shows monotonic RIS-aperture gains at $P = 15$ dBm: NOMA rises from 6.9 to 11.4 bps/Hz and OMA from 4.1 to 7.3 bps/Hz as N increases from 8 to 64.

For OMA, the learned variable is the time split $\mathbf{t} = [t_n, t_f, t_T]$. OMA–RIS adapts one shared RIS phase vector and uses per-user MRT; OMA no-RIS predicts only $\mathbf{t}$ from direct-link features.

Fig. 7 shows that NOMA-RIS gives the highest weighted rate over the tested power range and reaches near-zero QoS violation at lower power than the no-RIS and OMA references.

V. CONCLUSION

We studied joint power allocation and RIS-phase design for MISO NOMA-ISAC. The formulation combines cluster, individual-user, and sensing QoS constraints, while a first-order

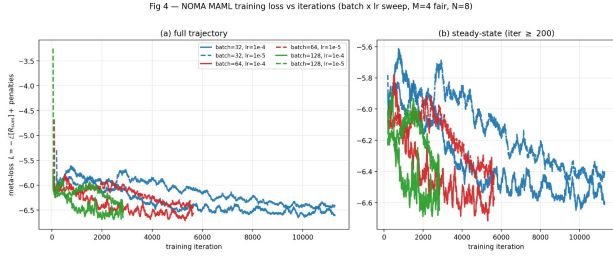
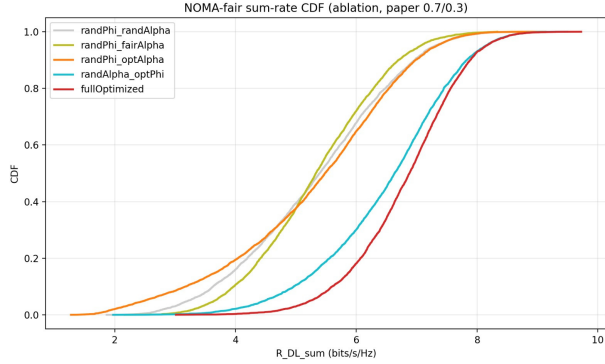
Fig. 3. MAML meta-loss: full trajectory and steady state ($M = 4$, $N = 8$).

Fig. 4. Weighted sum-rate CDF for the ablation configurations.

MAML solver learns the power map and adapts RIS phases per channel. Simulations show stable training and consistent gains from RIS assistance and NOMA across the evaluated operating points.

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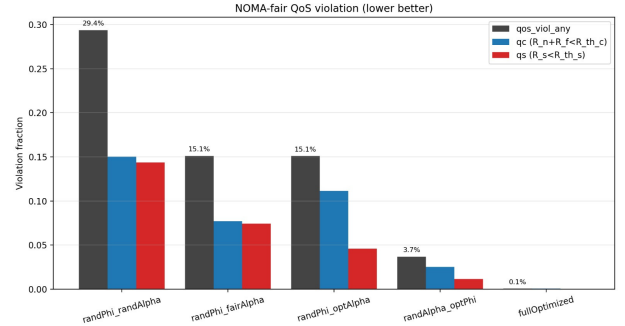
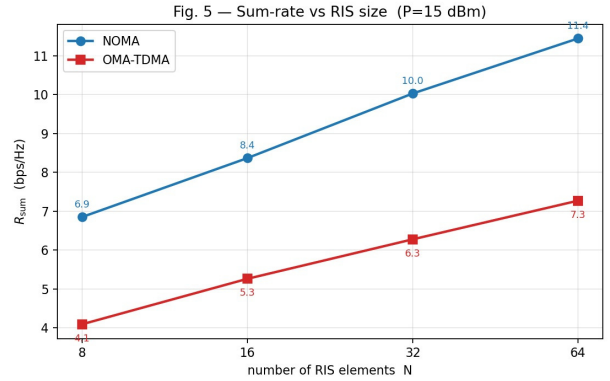


Fig. 5. QoS-violation fraction across the ablation.

Fig. 6. Weighted sum-rate versus RIS size at $P = 15$ dBm.

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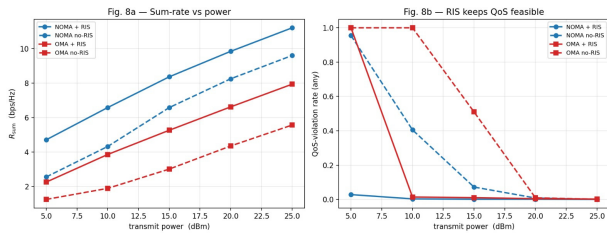


Fig. 7. Weighted rate and joint QoS violation versus transmit power.